

# CHAPTER 19

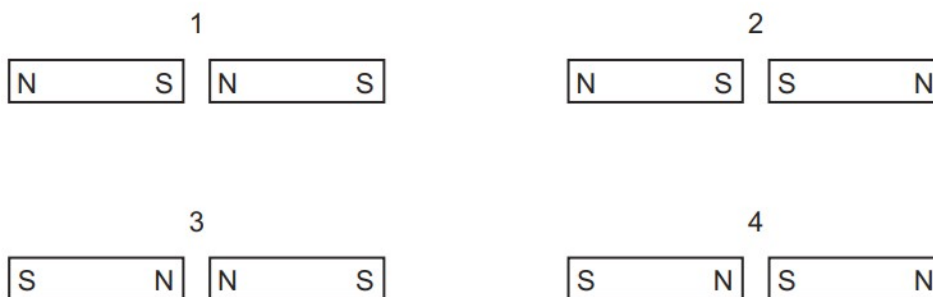
# SIMPLE MAGNETISM

## 4.1 Simple magnetism and magnetic fields

- 1 Describe the forces between magnetic poles and between magnets and magnetic materials, including the use of the terms north pole (N pole), south pole (S pole), attraction and repulsion, magnetised and unmagnetised
- 2 Describe induced magnetism
- 3 State the difference between magnetic and non-magnetic materials
- 4 State the differences between the properties of temporary magnets (made of soft iron) and the properties of permanent magnets (made of steel)
- 5 Describe a magnetic field as a region in which a magnetic pole experiences a force
- 6 Describe the plotting of magnetic field lines with a compass or iron filings and the use of a compass to determine the direction of the magnetic field
- 7 Draw the pattern and direction of the magnetic field lines around a bar magnet
- 8 State that the direction of the magnetic field at a point is the direction of the force on the N pole of a magnet at that point
- 9 Know that the relative strength of a magnetic field is represented by the spacing of the magnetic field lines
- 10 Describe uses of permanent magnets and electromagnets

**1. O/N/23/12/Q32**

Two permanent magnets are placed side by side in four different arrangements.



In which arrangements do the magnets repel each other?

- A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

**2. M/J/23/12/Q24**

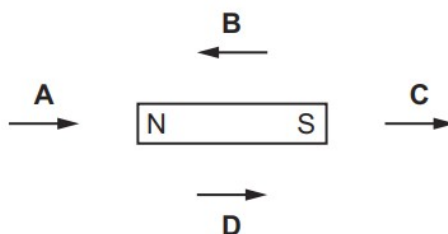
What proves that a metal bar is a permanent magnet?

- A** It attracts both ends of a compass needle.
- B** It attracts one end of another magnet.
- C** It conducts electricity.
- D** It repels one end of another magnet.

**3. M/J/23/12/Q25**

The diagram shows a bar magnet and four labelled arrows.

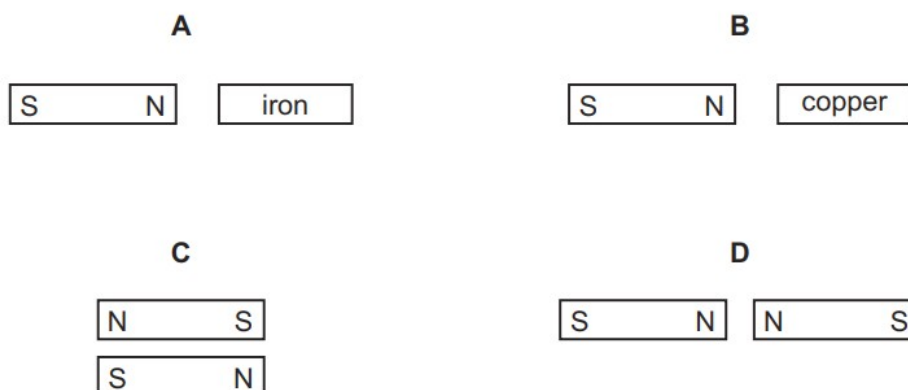
Which arrow shows the direction of the magnetic field at the point where the arrow is shown?



#### 4. M/J/23/11/Q25

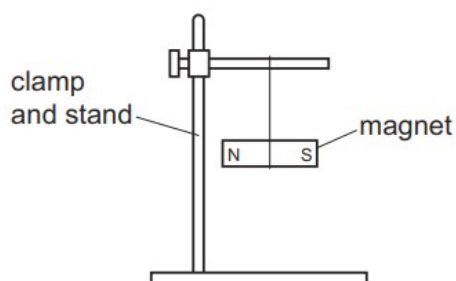
Bar magnets and various non-magnetic and demagnetised metal bars are placed in the different arrangements shown.

In which arrangement do the bars repel?



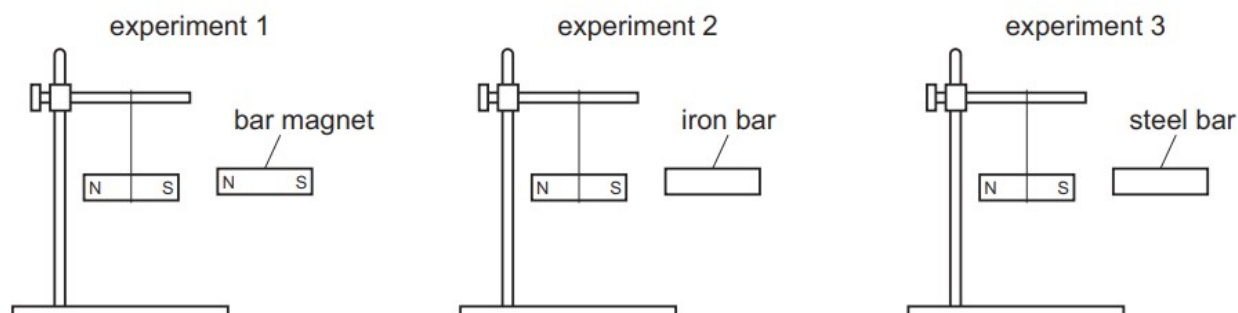
#### 5. M/J/23/11/Q26

A student uses a wooden clamp and a stand to hold a freely suspended bar magnet.



The diagrams show that:

- in experiment 1, the student holds a bar magnet next to the suspended magnet
- in experiment 2, the student holds an iron bar next to the suspended magnet
- in experiment 3, the student holds a steel bar next to the suspended magnet.



In which experiments does the suspended magnet move?

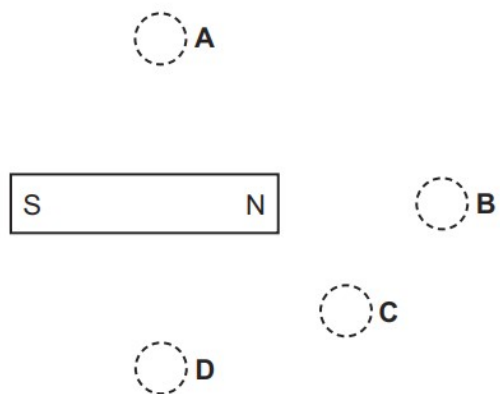
- A** 1, 2 and 3    **B** 1 and 2 only    **C** 1 and 3 only    **D** 2 and 3 only

**6. O/N/22/11/Q26**

Four plotting compasses are placed near a bar magnet. Ignore any effects of the Earth's magnetic field.

One compass appears like this .

What is a possible position for this compass?



**7. O/N/22/12/Q29**

The N pole of a permanent magnet is placed in turn close to a piece of iron, the S pole of another magnet, and a piece of copper.

Which objects are attracted to the N pole of the permanent magnet?

|          | piece of iron | S pole | piece of copper |
|----------|---------------|--------|-----------------|
| <b>A</b> | ✓             | ✓      | x               |
| <b>B</b> | x             | ✓      | x               |
| <b>C</b> | x             | ✓      | ✓               |
| <b>D</b> | x             | x      | ✓               |

key

✓ = attracted

x = not attracted

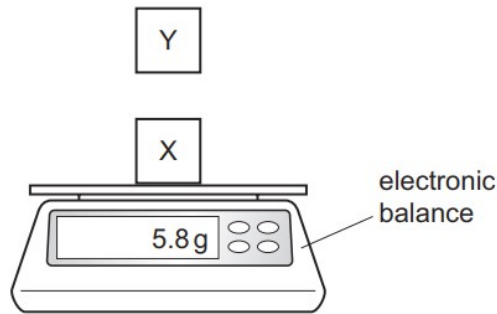
**8. M/J/22/11/Q27**

Which material is used for the core of an electromagnet?

- A** aluminium
- B** copper
- C** iron
- D** steel

**9. O/N/21/11/Q27**

Object X is on an electronic balance. Object Y is held above object X as shown.



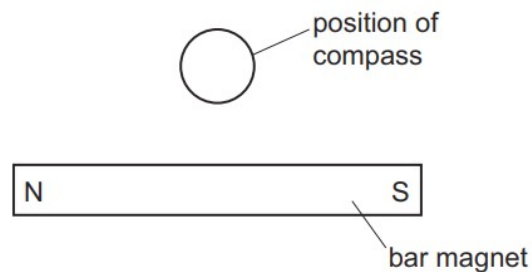
The reading on the balance increases when Y is moved closer to X.

Why does this happen?

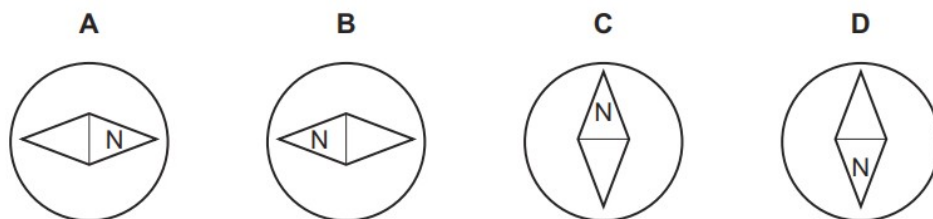
- A** X and Y are both made of iron.
- B** X and Y are both magnets.
- C** Y is a magnet and X is made of iron.
- D** Y is made of iron and X is a magnet.

**10. O/N/21/12/Q30**

A small plotting compass is placed near to a bar magnet as shown.



Which diagram shows the direction in which the compass needle points?



**11. M/J/21/12/Q30**

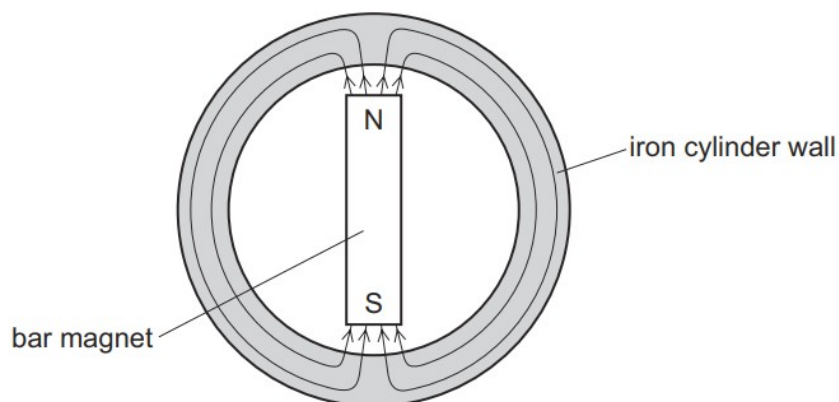
End X of a metal rod attracts the North pole of a compass needle.

Which statement about the rod is correct?

- A** It is made of copper that is not initially magnetised.
- B** It is made of copper with a South pole at X.
- C** It is made of steel that is not initially magnetised.
- D** It is made of steel with a North pole at X.

**12. O/N/20/11/Q32**

A bar magnet is placed in a hollow iron cylinder. The diagram shows the magnetic field pattern produced.

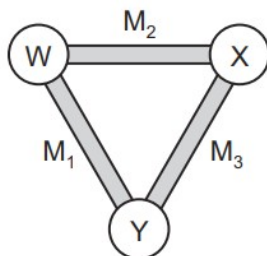


What is the cause of the field inside the cylinder wall?

- A** electromagnetic generation
- B** electromagnetic induction
- C** electrostatic induction
- D** magnetic induction

**13. O/N/19/11/Q30**

Magnets  $M_1$ ,  $M_2$  and  $M_3$  are joined by iron balls W, X and Y as shown.



Magnetic poles are induced in the iron balls W and X as shown.



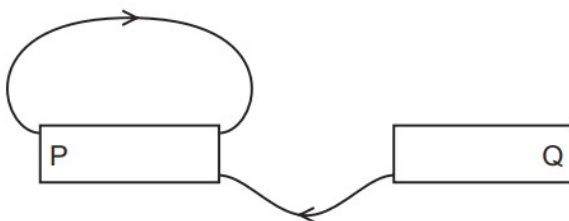
Which diagram shows the poles induced in ball Y?





**14. O/N/19/12/Q29**

The diagram shows two magnetic field lines of the magnetic field around two bar magnets.



Which row shows the magnetic poles at end P and end Q of the two magnets?

|          | P | Q |
|----------|---|---|
| <b>A</b> | N | N |
| <b>B</b> | N | S |
| <b>C</b> | S | N |
| <b>D</b> | S | S |

**15. M/J/19/12/Q31**

One end of each of two bars is marked with an X or a Y, as shown.



bar 1



bar 2

End X of bar 1 is brought close to the N-pole and then to the S-pole of a magnet.

End Y of bar 2 is brought close to the N-pole and then to the S-pole of the magnet.

The table shows what happens.

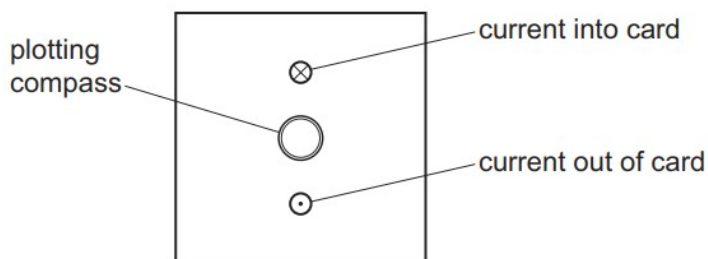
|                  | end X of bar 1 | end Y of bar 2 |
|------------------|----------------|----------------|
| N-pole of magnet | attracted      | attracted      |
| S-pole of magnet | attracted      | repelled       |

What materials are bar 1 and bar 2 made from?

|          | bar 1  | bar 2  |
|----------|--------|--------|
| <b>A</b> | copper | iron   |
| <b>B</b> | iron   | steel  |
| <b>C</b> | steel  | copper |
| <b>D</b> | steel  | iron   |

**16. M/J/19/11/Q29**

Two vertical wires pass at right-angles through a piece of card. There is a large current in each wire in the direction shown.



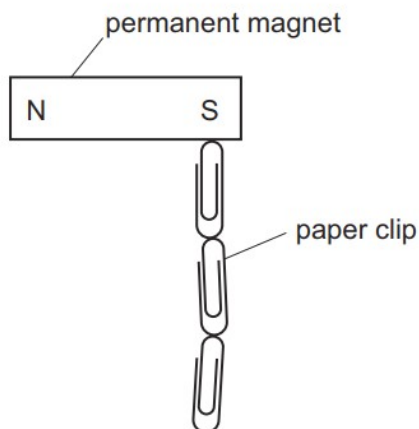
A plotting compass is placed on the card.

Which diagram shows the direction in which the needle of the plotting compass points?



**17. M/J/19/11/Q28**

A student investigates a permanent magnet by suspending paper clips from the magnet as shown.



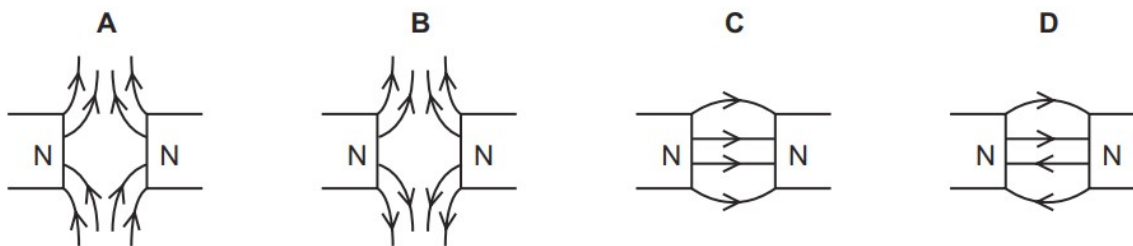
Which statement is correct?

- A** Each paper clip has a S-pole at the top and a N-pole at the bottom when suspended from the permanent magnet.
- B** If the paper clips are made of soft iron they become demagnetised when the permanent magnet is removed.
- C** If the paper clips are made of steel they become demagnetised when the permanent magnet is removed.
- D** More paper clips can be suspended from the N-pole.



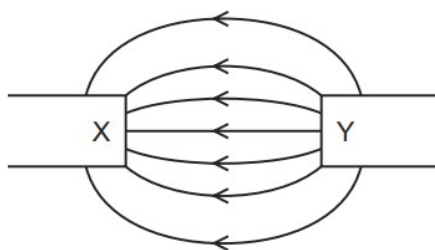
18. O/N/18/12/Q28

Which diagram shows the magnetic field pattern in the region between the N-poles of the two bar magnets?



19. O/N/17/11/Q26

The diagram shows the pattern and the direction of the magnetic field between two magnetic poles X and Y.



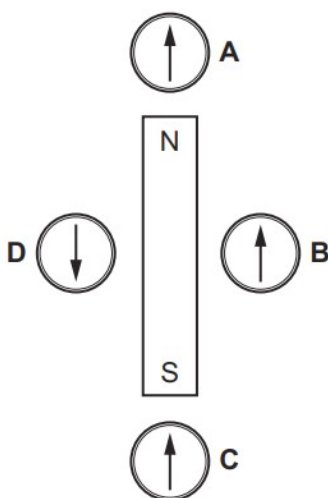
Which types of pole are X and Y?

|   | X      | Y      |
|---|--------|--------|
| A | N-pole | N-pole |
| B | N-pole | S-pole |
| C | S-pole | N-pole |
| D | S-pole | S-pole |

20. O/N/17/12/Q26

A compass needle is moved to different places around a strong bar magnet.

Which position shows **incorrectly** the direction of the compass needle?



**21. M/J/17/12/Q30**

What proves that a metal bar is a permanent magnet?

- A** It attracts both ends of a compass needle.
- B** It attracts one end of another magnet.
- C** It conducts electricity.
- D** It repels one end of another magnet.

**22. O/N/15/12/Q28**

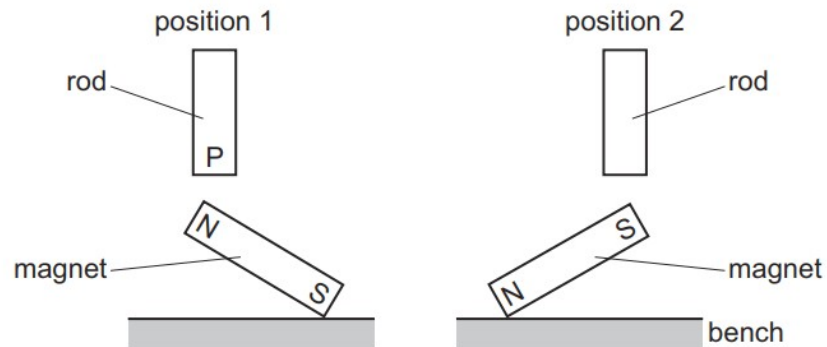
Which material can be picked up by a magnet?

- A** aluminium
- B** copper
- C** iron
- D** plastic

**23. M/J/15/11/Q25**

One end of a rod picks up the N-pole of a bar magnet when in position 1.

The same end of the rod picks up the S-pole of the bar magnet when in position 2.



Which material is the rod made from and what is the pole at end P of the rod when in position 1?

|          | material | pole at P |
|----------|----------|-----------|
| <b>A</b> | iron     | N-pole    |
| <b>B</b> | iron     | S-pole    |
| <b>C</b> | steel    | N-pole    |
| <b>D</b> | steel    | S-pole    |

**24. O/N/14/11/Q26**

A metal bar PQ hangs from a thin thread and always comes to rest with end P pointing north. Another bar XY of the same metal settles in no definite direction.

What happens if the two bars are brought near one another?

- A End P and end Q both attract end X.
- B End P attracts end X but repels end Y.
- C End P neither attracts nor repels end X.
- D End P repels end X but attracts end Y.

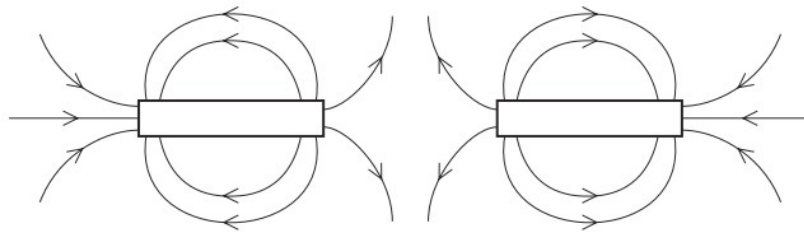
**25. M/J/14/12/Q30**

What always produces a permanent bar magnet?

- A an iron bar in a coil carrying alternating current (a.c.)
- B an iron bar in a coil carrying direct current (d.c.)
- C a steel bar in a coil carrying alternating current (a.c.)
- D a steel bar in a coil carrying direct current (d.c.)

**26. M/J/13/12/Q29**

The magnetic field around two bar magnets is shown.

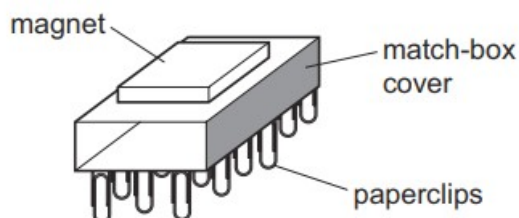


Which diagram represents the correct arrangement of magnetic poles?

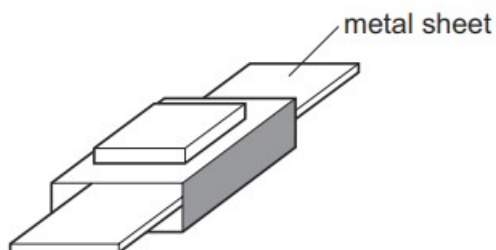
- |   |                                                                                           |                                                                                           |
|---|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| A | <div style="border: 1px solid black; padding: 2px; display: inline-block;">N      S</div> | <div style="border: 1px solid black; padding: 2px; display: inline-block;">N      S</div> |
| B | <div style="border: 1px solid black; padding: 2px; display: inline-block;">N      S</div> | <div style="border: 1px solid black; padding: 2px; display: inline-block;">S      N</div> |
| C | <div style="border: 1px solid black; padding: 2px; display: inline-block;">S      N</div> | <div style="border: 1px solid black; padding: 2px; display: inline-block;">N      S</div> |
| D | <div style="border: 1px solid black; padding: 2px; display: inline-block;">S      N</div> | <div style="border: 1px solid black; padding: 2px; display: inline-block;">S      N</div> |

**27. M/J/13/11/Q26**

A teacher sticks a magnet to the top surface of a match-box cover. The bottom surface is placed in a small tray of iron paperclips. As the match-box cover is lifted up, a large number of paperclips are held on the bottom surface.



Sheets of metal are placed inside the match-box cover, between the magnet and the paperclips.



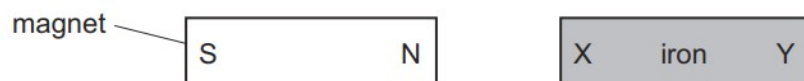
When sheet X is placed inside, the paperclips stay attached. When sheet Y is placed inside, all the paperclips fall off.

Which metals are the sheets made from?

|          | sheet X   | sheet Y   |
|----------|-----------|-----------|
| <b>A</b> | aluminium | copper    |
| <b>B</b> | copper    | iron      |
| <b>C</b> | iron      | aluminium |
| <b>D</b> | iron      | copper    |

**28. O/N/12/12/Q26**

A magnet is placed close to an iron bar.



The iron bar becomes an induced magnet.

Which magnetic poles are formed at the ends X and Y of the bar?

|          | end X | end Y |
|----------|-------|-------|
| <b>A</b> | N     | N     |
| <b>B</b> | N     | S     |
| <b>C</b> | S     | N     |
| <b>D</b> | S     | S     |

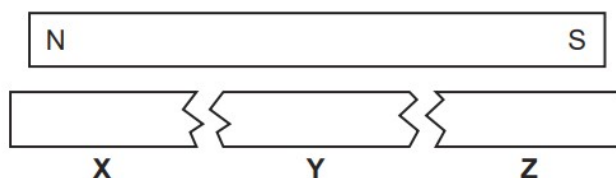
**29. M/J/11/12/Q25**

Which list contains an example of a non-magnetic material, a magnetic material and a magnetised material?

- A copper, iron, a compass needle
- B copper, iron, polythene
- C iron, steel, a compass needle
- D iron, steel, polythene

**30. O/N/04/Q27**

A bar magnet is broken into three parts **X**, **Y** and **Z**.



Which diagram shows the poles in **X**, **Y** and **Z**?

- A 

|   |   |   |
|---|---|---|
| X | Y | Z |
| N |   | S |
- B 

|   |   |   |   |
|---|---|---|---|
| N | N | S | S |
|---|---|---|---|
- C 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| N | N | S | N | S | S |
|---|---|---|---|---|---|
- D 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| N | S | N | S | N | S |
|---|---|---|---|---|---|